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Scale-dependent contraction of spatial wet-bulb temperature contrasts in eastern China

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Abstract

Regional summaries of wet-bulb temperature omit the spatial distribution of humid heat. This study examines whether geographic contrast changes when the regional mean is high. Spatial contrast across 121 eastern China sites is measured with Gaussian-weighted graph Laplacians and summarised over five log-spaced scales; product cyclic shifts provide finite-record assessment. The statistic and field-selection rules were developed using the 2015 and 2022 summers and applied to 33 other summers in 1991–2025. High days had 7.28% lower spatial contrast than middle days, and 28 of 33 summer estimates were negative. An exploratory 99,999-draw product cyclic-shift analysis gave $p = 0.00001$ under the conditional product-invariance null. Contraction increased from 2.86% at 126 km to 13.27% at 2,013 km. An exact Laplacian decomposition attributed the broad-scale change chiefly to high-day anomalies opposing monthly climatological patterns; transient-anomaly energy changed little. The same scale pattern appeared in 1950–1990 (11.04% mean contraction; 40 of 41 negative summers). Station data from ten non-development summers yielded concordant broad-scale contraction (17.34%; nine of ten negative). High regional humid heat is associated with a flatter broad-scale field, a feature the regional mean cannot identify.

Keywords ERA5-Land, graph Laplacian, humid heat, randomisation inference, semivariance, spatial statistics

1. Introduction

Wet-bulb temperature (WBT) is a widely used measure of humid heat stress that combines air temperature and humidity (Sherwood and Huber, 2010; Raymond et al., 2020). Two days can have identical regional means yet sharply different spatial fields: one may contain strong north–south or local contrasts, while the other is comparatively coherent. The analysis therefore focuses on spatial contrast in the WBT field itself.

Recent work has documented increasingly concentrated humid-heat extremes in standard climate regions (Speizer et al., 2022), sharp local wet-bulb extremes associated with wet soils (Chagnaud et al., 2025), and reduced intra-urban heat-stress variability during some heat waves (Shreevastava et al., 2023). Spatial-extremes models such as Healy et al. (2025) estimate nonstationary temperature tails and the area exceeding critical thresholds. The within-month comparison here examines how pairwise WBT differences change with graph scale. A regional mean discards spatial arrangement, while ordinary cross-site variance cannot distinguish neighbouring contrasts, mesoscale structure and a domain-wide gradient. Empirical variograms represent distance-dependent variation, but comparing regimes requires a common summary across distance scales. Smooth weighting provides that summary and permits an exact spatial allocation of the regional contrast.

We ask whether high regional mean wet-bulb temperature in eastern China is accompanied by a contraction of spatial differences. The question is associational. A negative contrast is compatible with broad atmospheric forcing, but topography (Pepin et al., 2015), soil moisture (Seneviratne et al., 2010), circulation (Horton et al., 2015), and measurement support can produce or mask similar patterns. If event labels depend on spatial dispersion as well as the regional mean, the outcome influences which days are selected, and a difference can be created by construction (Simmons et al., 2011). If the most favourable distance is chosen after inspecting the data, scale-specific evidence is likewise overstated (White, 2000). Event definitions therefore use regional mean wet-bulb temperature, and the spatial scales were selected during method development.

We combine established graph Dirichlet energies into a scale-indexed regime contrast for observed spatial fields. Under second-order stationarity, each energy is a smoothly weighted semivariance (Cressie, 1993; Shuman et al., 2013; Dong et al., 2016). The construction averages record-specific relative contrasts over five log-spaced graphs, preserves the daily multiscale profile in product cyclic shifts, and decomposes the regional effect exactly over nodes and low-dimensional spatial components. This combination reveals which geographic scales and structures account for a change in WBT contrast. Ordinary variance discards distance, binned semivariances impose hard boundaries, and Moran's I and Geary's C standardise away the amplitude of spatial differences.

The application uses 2015 and 2022 for method development and the other 33 summers for evaluation. A product cyclic-shift calculation assesses the multiscale statistic under a conditional product-invariance null. Equal edge splitting and Laplacian-orthogonal projections distinguish attenuation of persistent geographic structure from changes in transient anomalies.

2. Multiscale graph regime contrasts

2.1. Scale-indexed graph dispersion and regime contrast

Let y_{it} denote WBT at location i in the selected field for day t, and let $\bar{y}_t = n^{-1}\sum_i y_{it}$. The benchmark is the site mean squared spatial deviation,

$$V_t = \frac{1}{n}\sum_{i=1}^{n}(y_{it} - \bar{y}_t)^2, \qquad n = 121. \tag{1}$$

To retain spatial arrangement, we use an equirectangular kilometre projection centred at the mean site latitude: longitude is multiplied by $111.32\cos(\bar{\phi})$ and latitude by 110.57. For bandwidth h, set $w_{ij,h} = \exp\{-d_{ij}^2/(2h^2)\}$ for $i \neq j$ and $w_{ii,h} = 0$. Let $\mathbf{W}_h = (w_{ij,h})$, $\mathbf{L}_h = \mathrm{diag}(\mathbf{W}_h\mathbf{1}) - \mathbf{W}_h$, and $S_h = \sum_{i<j} w_{ij,h}$. We define graph dispersion as

$$Q_h(\mathbf{y}) = \frac{\mathbf{y}^\top\mathbf{L}_h\mathbf{y}}{2S_h} = \frac{\sum_{i<j} w_{ij,h}(y_i - y_j)^2}{2\sum_{i<j} w_{ij,h}}. \tag{2}$$

Graph dispersion is therefore an edge-weighted average of pairwise squared differences. Dividing by S_h makes the measure comparable across graphs with different total edge weights. Smaller values indicate greater graph-scale coherence. Graph dispersion measures spatial contrast, distinct from threshold-exceedance area. It has units $^\circ\mathrm{C}^2$; $D_h^{\mathrm{RMS}} = (2Q_h)^{1/2}$ is the weighted root mean squared pairwise WBT difference in $^\circ\mathrm{C}$.

Equation (2) connects three familiar descriptions of spatial variation. It is the graph Dirichlet energy used to measure graph signal roughness (Shuman et al., 2013; Dong et al., 2016). If the field is second-order stationary with semivariogram $\gamma(d_{ij}) = \frac{1}{2}\,\mathrm{E}(Y_i - Y_j)^2$, then

$$\mathrm{E}\{Q_h(\mathbf{Y})\} = \frac{\sum_{i<j} w_{ij,h}\gamma(d_{ij})}{S_h}, \tag{3}$$

so its expected profile is a kernel average of the semivariogram. This interpretation requires second-order stationarity; the finite-field statistic does not. In the nonstationary WBT fields studied here, Q_h is interpreted only as graph-scale dispersion. On a complete graph with equal off-diagonal weights, $Q_h = nV/(n-1)$. Ordinary spatial variance is therefore the complete-graph endpoint of the same family, up to its degrees-of-freedom factor.

We use five bandwidths selected during method development: $h/m \in \{0.125, 0.25, 0.5, 1, 2\}$, where m is the median pairwise distance. They correspond to 126, 252, 503, 1,006 and 2,013 km. Their weighted mean pair distances are 200, 319, 549, 826 and 992 km, and their effective edge counts are 366, 1,138, 3,223, 6,005 and 7,109 out of 7,260. Thus 126 km denotes a Gaussian bandwidth with no hard distance cut-off; it is local only relative to the sampled support, where the median nearest-neighbour distance is 150 km. For edge weights w_e, the reported Kish quantity is $N_{\text{eff}} = (\sum_e w_e)^2 / \sum_e w_e^2$.

Classifying days separately within each month-year record $r = (y, m)$ prevents the seasonal cycle or an interannual mean shift from entering the event definition. Let $q_{25,r}$ and $q_{75,r}$ be the type-7 sample quartiles of $\bar{y}_t$ in record r. A day is labelled high if $\bar{y}_t \geq q_{75,r}$ and middle if $q_{25,r} \leq \bar{y}_t < q_{75,r}$. Lower-quartile days do not enter the contrast. There were no threshold ties; every month-year contains eight high days and 14 or 15 middle days. These are relative upper-quartile days: the threshold is redefined within each record, and no extreme-value tail is involved.

The two groups are compared at each scale by the relative difference of their dispersion means. Let $\mathcal{E}_{ym}$ and $\mathcal{M}_{ym}$ be the high- and middle-day sets in record (y, m), and let $\overline{Q}^{E}_{ymh}$ and $\overline{Q}^{M}_{ymh}$ denote the corresponding graph-dispersion means. The scale-specific relative contrast and its summer average are

$$R_{ymh} = \frac{\overline{Q}^{E}_{ymh}}{\overline{Q}^{M}_{ymh}} - 1, \qquad \widehat{\theta}_y = \frac{1}{3H} \sum_{m=1}^{3} \sum_{h=1}^{H} R_{ymh}. \tag{4}$$

For a set $\mathcal{Y}$ of summers, the finite-record estimand is

$$\widehat{\theta}_{\mathcal{Y}} = \frac{1}{|\mathcal{Y}|} \sum_{y \in \mathcal{Y}} \widehat{\theta}_y. \tag{5}$$

A negative value means that high-regional-mean fields vary less across space. Because the bandwidths are equally spaced on the log scale, the arithmetic mean gives equal weight to every bandwidth in the five-scale grid. The log-ratio analysis replaces R_{ymh} by $\log(\overline{Q}^{E}_{ymh} / \overline{Q}^{M}_{ymh})$ at the same three stages of averaging and back-transforms the final mean for interpretation. A second analysis uses the bounded symmetric contrast $2(\overline{Q}^{E}_{ymh} - \overline{Q}^{M}_{ymh}) / (\overline{Q}^{E}_{ymh} + \overline{Q}^{M}_{ymh})$. The three effect measures are applied to identical fields and labels in the denominator stress test.

2.2. Exact spatial and structural decomposition

Graph dispersion accumulates over edges. We obtain an exact and symmetric site allocation by splitting every undirected edge equally between its endpoints, although other allocations are possible. For location i, define

$$q_{ih}(\mathbf{y}) = \frac{1}{4S_h} \sum_{j \neq i} w_{ij,h} (y_i - y_j)^2. \tag{6}$$

The identity is exact because the double sum counts every edge twice:

$$\sum_{i=1}^{n} q_{ih}(\mathbf{y}) = \frac{1}{2S_h} \sum_{i<j} w_{ij,h} (y_i - y_j)^2 = Q_h(\mathbf{y}). \tag{7}$$

The allocation is translation invariant and nonnegative for a single field. It also respects the graph geometry: a site receives contributions only from its incident weighted contrasts, while distant pairs carry little weight in a local graph. Equal edge splitting avoids assigning an arbitrary direction to an undirected spatial relationship.

Let $\overline{q}^{E}_{iymh}$ and $\overline{q}^{M}_{iymh}$ denote the high- and middle-day means within record (y, m), and let $\overline{Q}^{M}_{ymh} = \sum_i \overline{q}^{M}_{iymh}$. The site contribution to the record-level relative effect is

$$c_{iymh} = \frac{\overline{q}^{E}_{iymh} - \overline{q}^{M}_{iymh}}{\overline{Q}^{M}_{ymh}}, \qquad \sum_i c_{iymh} = R_{ymh}. \tag{8}$$

Unlike q_{ih}, c_{iymh} may be negative because it measures a change between event regimes. Its denominator is the record-specific middle-day dispersion. Sites in months with different baseline variability therefore enter the final allocation on the same relative scale as the corresponding record-level estimator.

Let $\mathcal{Y}$ be the set of summers. With $H = 5$, the reported node allocation is

$$c_i = \frac{1}{|\mathcal{Y}|} \sum_{y \in \mathcal{Y}} \frac{1}{3} \sum_{m=1}^{3} \frac{1}{H} \sum_{h=1}^{H} c_{iymh}, \qquad \sum_{i=1}^{n} c_i = \widehat{\theta}_{\mathcal{Y}}. \tag{9}$$

The order and weights in Equation (9) match the estimator: equal months within a summer, equal graph scales, and equal summers. The identity is checked numerically after every analysis. Changing the spatial support changes the incident edges and hence the allocation. Accordingly, every main-text map is calculated from the 121-node graph. For a scale-specific map, we retain one h in Equation (9) instead of averaging over H; the observed-node sum then recovers the corresponding scale-specific contrast.

For display, we fit a low-rank thin-plate REML surface to each set of 121 observed-node values and evaluate it on a fine land-masked raster. Graph construction, estimation, uncertainty calculations and node-sum identities all use the observed nodes, which are overlaid on each map. A negative c_i allocates part of the regional contraction to edges incident to site i; the allocation is descriptive at the coordinate level. The maps also show regime means of $y_{it} - \bar{y}_t$ and their difference. These spatially centred fields separate a change in spatial pattern from a shift in the daily regional mean.

To quantify the dominant north–south structure, let $\boldsymbol{\ell}$ be centred latitude and $\mathbf{z}_t = \mathbf{y}_t - \bar{y}_t \mathbf{1}$. Define

$$\widehat{\beta}_{th} = \frac{\boldsymbol{\ell}^\top \mathbf{L}_h \mathbf{z}_t}{\boldsymbol{\ell}^\top \mathbf{L}_h \boldsymbol{\ell}}, \qquad \mathbf{e}_{th} = \mathbf{z}_t - \widehat{\beta}_{th} \boldsymbol{\ell}.$$

Orthogonality in the $\mathbf{L}_h$ inner product gives

$$Q_h(\mathbf{y}_t) = \frac{\widehat{\beta}_{th}^2 \boldsymbol{\ell}^\top \mathbf{L}_h \boldsymbol{\ell}}{2S_h} + \frac{\mathbf{e}_{th}^\top \mathbf{L}_h \mathbf{e}_{th}}{2S_h}. \tag{10}$$

High-minus-middle means of the two terms, divided by middle-day total dispersion, add exactly to R_{ymh}. This exact energy accounting is descriptive. More generally, for a centred spatial basis matrix $\mathbf{B}$, the Laplacian projection has coefficients $\widehat{\boldsymbol{\beta}}_{th} = (\mathbf{B}^\top \mathbf{L}_h \mathbf{B})^- \mathbf{B}^\top \mathbf{L}_h \mathbf{z}_t$, where $(\cdot)^-$ is a generalised inverse. Centred latitude is the reference structural summary. Further projections add longitude and then elevation, obtained by dividing the official invariant ERA5-Land geopotential field by standard gravity. We compare the total structured and residual energies of these nested column spaces; no order-specific latitude, longitude or elevation contribution is assigned.

A second decomposition separates the persistent monthly field from daily departures. Write $\mathbf{y}_t = \boldsymbol{\mu}_m + \mathbf{a}_t$, where $\boldsymbol{\mu}_m$ is the site-specific 35-summer mean field for month m. Then

$$Q_h(\mathbf{y}_t) = \frac{\boldsymbol{\mu}_m^\top \mathbf{L}_h \boldsymbol{\mu}_m}{2S_h} + \frac{\boldsymbol{\mu}_m^\top \mathbf{L}_h \mathbf{a}_t}{S_h} + \frac{\mathbf{a}_t^\top \mathbf{L}_h \mathbf{a}_t}{2S_h}. \tag{11}$$

The first term is constant within a month and cancels from the high-minus-middle numerator. The other two terms quantify change in anomaly energy and change in climatology–anomaly alignment. Dividing both differences by the middle-day total dispersion yields two components that add exactly to R_{ymh}.

3. Finite-record inference and study design

3.1. Product cyclic-shift randomisation

Days within a month are serially dependent, so each month-year record is randomised as a unit. Its complete daily outcome profile is rotated by a randomly chosen offset while the mean-WBT series and labels remain in place. This joint rotation preserves dependence between bandwidths, and the statistic is recomputed after every shift. The one-sided Monte Carlo p-value is

$$p = \frac{1 + \sum_{b=1}^{B} I(T_b^{\text{shift}} \le T_{\text{obs}})}{B+1}. \tag{12}$$

Validity requires cyclic invariance, a stronger property than generic stationarity. For record r, let $X_r = (\bar{y}_{rt})_{t=1}^{n_r}$ denote the mean series and let D_r collect the H graph-dispersion values for each day, with $X = (X_1, \ldots, X_R)$ and

$D = (D_1, \ldots, D_R)$. The operator C_{n_r} rotates all H columns of D_r together, and $G = \prod_r C_{n_r}$ is the group of all joint rotations. The null hypothesis is group invariance:

$$(D \mid X) \overset{d}{=} (gD \mid X) \quad \text{for every } g \in G. \tag{13}$$

Proposition 1 (Product cyclic-shift validity) *Let G be finite and suppose Equation (13) holds. For a measurable lower-tail statistic $T(X, D)$ whose definition is fixed before the transformations are examined, define*

$$p_G(X, D) = \frac{1}{|G|} \sum_{g \in G} I\{T(X, gD) \le T(X, D)\}.$$

Then, for every $u \in [0, 1]$,

$$\Pr\{p_G(X, D) \le u \mid X\} \le u \quad \text{almost surely.}$$

If the $|G|$ transformed statistics have no ties, p_G is uniform on $\{1/|G|, \ldots, 1\}$ conditional on the invariant orbit, and the enumeration test is exact at its attainable levels. If B transformations are sampled independently and uniformly from G, independently of (X, D), the plus-one value p_B in Equation (12) satisfies

$$\Pr\{p_B \le u \mid X\} \le \frac{\lfloor (B+1)u \rfloor}{B+1} \le u.$$

Proposition 1 is a finite-sample result: no independence between days is needed. The conditional statement holds even with ties or repeated group actions, although exact uniformity on the attainable grid requires the no-tie condition stated above. The full proof is in Supplementary Section S1, and the Monte Carlo version follows the argument of Phipson and Smyth (2010).

Truncated time-shift tests replace wrap-around invariance with strict stationarity (Harris, 2020; Yuan and Shou, 2024), but their conservative finite-sample form requires at least 19 shifts on each side for a 0.05 cutoff. Records of 30 or 31 days leave no usable aligned window under that requirement, so we use cyclic shifts under Equation (13).

The product condition is stronger than marginal cyclic invariance of each record because it permits the records to be rotated independently. A targeted stress simulation in Supplementary Section S11.3 examines a shared cross-record temporal process whose positive loadings violate this joint product-invariance condition.

For scale-specific inference in the development records, each shifted profile contributes its minimum relative effect across the five bandwidths, and an observed scale contrast is compared with the distribution of these minima. Under the complete joint null, this single-step comparison controls the family-wise error rate in the weak sense; strong control need not hold under arbitrary partial nulls. Because every component is a unitless relative change, taking the minimum does not mix quantities measured on different scales.

A second uncertainty summary is built from the six month-year contrasts: their unweighted mean, with a 95% interval formed from a t_5 critical value and the between-record standard error. This interval is a descriptive summary of variation among the six records.

3.2. Development and evaluation design

The summers of 2015 and 2022 served as the design sample; the other 33 summers form the evaluation record. Event labels, bandwidths and field selection from the design stage carry into Equations (4)– (5). The target $\widehat{\theta}_{\mathcal{Y}_H}$ describes the evaluation record, without a stationarity assumption across years.

For the global cyclic-shift analysis, set $T_{\text{obs}} = \widehat{\theta}_{\mathcal{Y}_H}$. Each Monte Carlo transformation rotates the five-column dispersion profile within each of the 99 evaluation month-year records, then recomputes every record ratio and the equal-month, equal-scale and equal-summer mean, with $B = 99{,}999$.

We added the global product cyclic-shift distribution after estimating the 33-summer contrast and therefore report its p-value as exploratory. The statistic is the evaluation-record estimand in Equation (5).

Effect size is reported through the record mean, summer-specific contrasts, negative-summer count and leave-one-summer-out range. Supplementary Table S1 summarises the analysis sequence and the role of each calculation.

3.3. Process-level reference

For comparison, a process-level reference treats the summer sequence as a stationary weakly dependent process. Under standard strong-mixing, moment and long-run variance conditions, its mean admits a central-limit approximation; a growing-bandwidth HAC estimator needs a stronger uniform autocovariance condition (Ibragimov and Linnik, 1971; Newey and West, 1987; Andrews, 1991). The full statement, ratio negative-moment conditions and proof are in the Supplement. At 33 summers, simulation shows that Student and HAC intervals can have substantial undercoverage under serial dependence. Both intervals are reported alongside the finite-record randomisation result.

4. Simulation study

The simulations examine day-level cyclic randomisation and the behaviour of process-model intervals at the 33-summer record length. These are separate sampling problems: additional days within a month do not increase the number of summers. Table 1 specifies both designs.

4.1. Data-generating processes

The day-level experiment uses the observed coordinates and six 30–31-day records. Its circular Gaussian null satisfies Proposition 1; two finite-autoregression variants deliberately violate wrap-around invariance. Each Gaussian anomaly is projected through $\mathbf{I} - \mathbf{1}\mathbf{1}^\top/n$, so its sample mean across the 121 sites is exactly zero and the field mean equals the series used to form labels. The centred anomaly is independent of that mean. This calibration design isolates the randomisation properties; the fixed-hour and anomaly-field analyses in Section 5 examine field selection and label–field dependence in the observed record. Amplitude reduction, correlation-range expansion and gradient suppression produce true profile contractions of 5.9%–27.4%, calculated from the known quadratic forms. The Monte Carlo standard error is 0.7 percentage points at 5% rejection and 1.6 points at 50% power, so differences of one or two points are difficult to distinguish.

Seven procedures are compared in the day-level experiment: the mean of five graph-dispersion contrasts, ordinary spatial variance, a four-nearest-neighbour semivariance, a classical binned semivariance using pairs 400–600 km apart (Cressie, 1993), an equal-weight profile of five semivariance contrasts formed from pair-distance quintiles, and global Moran's I and Geary's C (Moran, 1950; Geary, 1954) with the 503-km Gaussian weights. The five dispersion quantities use lower-tail tests; because the direction of an autocorrelation change depends on the mechanism, Moran and Geary use the absolute high-minus-middle difference and a two-sided group test. The single distance bin is a local benchmark; the five-bin profile is the closest variogram-profile analogue. Specifically, the 7,260 pair distances are split at their quintiles into sets $\mathcal{B}_k$, and

$$\Gamma_k(\mathbf{y}) = \frac{1}{2|\mathcal{B}_k|} \sum_{(i,j)\in\mathcal{B}_k} (y_i - y_j)^2.$$

The profile's statistic applies Equation (4) to each Γ_k, with the same equal-record, equal-bin averaging as the graph statistic. All metrics are evaluated on identical simulated fields with the same joint shifts. The graph profile avoids hard bin boundaries and admits the exact node allocation in Equation (9); we compare calibration and power to assess whether these properties entail a loss relative to the other six procedures.

The repeated-summer experiment compares the Student interval, fixed lag-2 calculation, Bartlett estimator with $L_R = \lfloor R^{1/3} \rfloor$, sign test and three-test rule. The -7% alternative is close to the evaluation-record estimate.

4.2. Calibration and comparison across spatial mechanisms

Table 2 reports rejection rates over the nine spatial alternatives; Figure 1 shows how each method's power range depends on the unknown mechanism.

Across eight cyclic-invariant designs, graph-profile rejection ranged from 4.5% to 6.8%, while the six competitors spanned 4.0%–6.4%. The graph test had rejection rates of 4.5% at temporal correlation 0.75 and 6.1% under the t_3 field. The two non-circular boundary cases gave 4.6% and 5.1%.

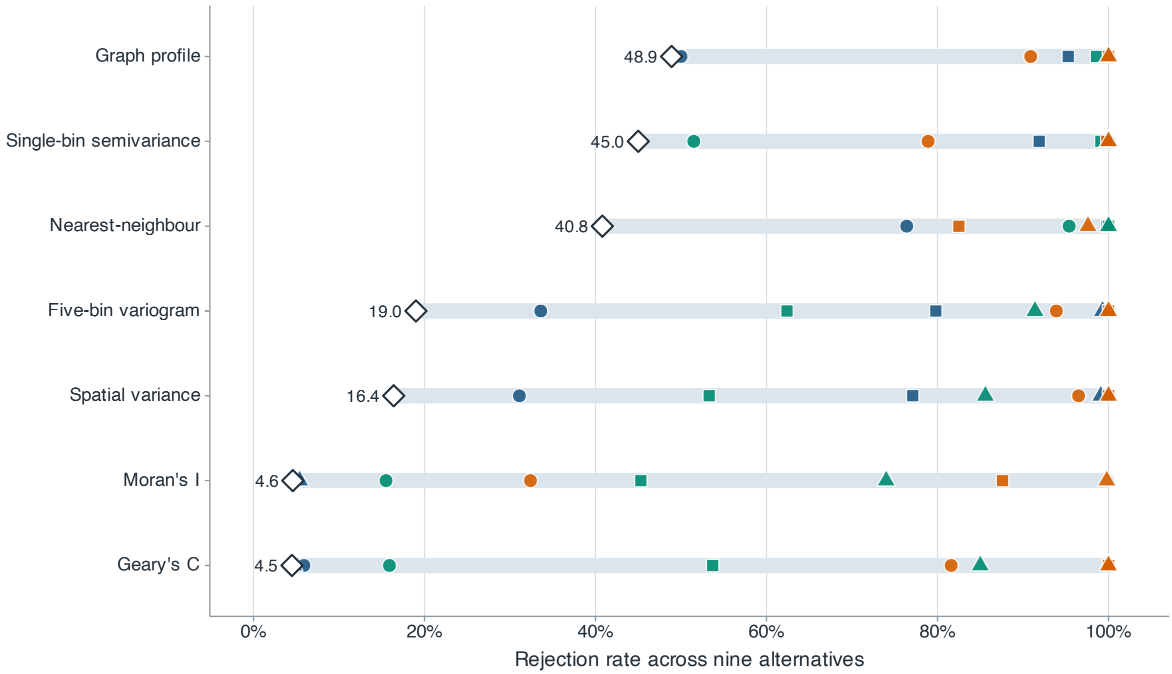

Figure 1 Power across nine simulated spatial alternatives. Blue, green and orange marks denote amplitude reduction, correlation-range expansion and gradient suppression; circles, squares and triangles denote weak, medium and strong changes. For each method, the pale segment spans power across the nine alternatives and the diamond marks the minimum.

Gaussian-alternative absolute bias was below 0.48 percentage points and root mean squared error was 2.6–3.8 points. Under the original t_3 null, graph randomisation size was 6.1%. A paired denominator stress test then used 2,000 heavy-tailed data sets under each of the null and -7% alternatives and 999 common product shifts per data set. Null rejection was 4.70%, 4.90% and 4.75% for the raw ratio, log ratio and bounded symmetric contrast. On their native scales, the corresponding null absolute biases were 0.372, 0.070 and 0.066, and root mean squared errors were 1.472, 0.328 and 0.277. Under the -7% alternative, root mean squared errors were 1.369, 0.328 and 0.275, while rejection was 7.95%, 8.50% and 7.90%. The bounded contrast sharply reduced estimator dispersion but did not materially increase power in this design. Randomisation calibration can therefore coexist with unstable effect estimation; in this design, resistant summaries reduce but do not eliminate the problem.

Performance varies across the mechanisms in Table 2. Nearest-neighbour semivariance is strongest for range changes, but its power is 40.8% under weak gradient suppression, compared with 90.9% for the graph profile and 96.5% for variance. The five-bin variogram is similar to the graph profile for gradient suppression (93.9% versus 90.9% at the weakest setting), but is less sensitive to the weak amplitude and range changes (33.6% and 19.0% versus 50.0% and 48.9%). Moran's I and Geary's C have near-nominal power for pure amplitude changes because standardisation removes amplitude. The profile's minimum power across the nine tabulated alternatives is 48.9%, higher than the minimum of each other procedure in this experiment.

The scale-specific profiles retain information lost by a scalar test. Strong amplitude reduction changes all three summaries by about 19%; range expansion acts most strongly on the local graph (28.2%), while gradient suppression acts on variance and the broad graph (43.9% and 42.5%) but changes the local graph by only 9.3%.

Eight additional scenarios allow the label-generating mean and spatial field to share structure; full specifications are in the Supplement. In the shared-latent null, the sign of the spatial gradient has mean correlation 0.61 with regional WBT, but its squared energy is constant, so the true dispersion contrast is zero. The other nulls add a within-month trend, anisotropic spatial covariance or selection of a daily peak from 24 simulated hourly regional means, with hour-invariant anomaly covariance. Corresponding alternatives attenuate the gradient, impose a seasonally declining gradient or reduce high-day anomaly amplitude. Graph null rejection is 4.0%–7.0%, compared with 4.3%–7.3% for the five-bin variogram. Graph rejection under these four nulls is close to its cyclic-invariance calibration. The graph profile is at least as powerful in three of four alternatives, whereas the variogram is slightly stronger for the seasonal-gradient alternative. Smooth graph weighting and exact allocation therefore show no consistent power penalty across these scenarios.

4.3. Process-reference stress test

At $R = 33$, increasing Gaussian year correlation from 0 to 0.6 reduces Student coverage from 95.2% to 66.1%, fixed-lag coverage from 93.3% to 78.7%, and growing-HAC coverage from 91.2% to 79.0%. Corresponding Gaussian null rejection rises to 21.2%, 14.4% and 14.1%. The error arises mainly from long-run variance estimation. Coverage from all three process-model intervals deteriorates at this record length under the simulated dependence. Full results for 108 cells, including skewed and heavy-tailed innovations, appear in the Supplement.

An additional 10,000-replication experiment uses the exact 1991–2025 calendar and removes 2015 and 2022. The largest absolute mean bias over correlations 0, 0.3 and 0.6 and effects 0 and -7% is 0.03 percentage points. Excluding the two design summers therefore has negligible first-order centring bias, but short-record dependence remains: at correlation 0.6, null coverage is 68.3% for the independent-summer interval and 80.2% for calendar lag 2.

5. Humid-heat application

We apply graph dispersion to synchronous wet-bulb temperature fields in eastern China during 1991–2025 and extend the analysis to 1950–1990.

5.1. Data construction and analysis design

ERA5-Land supplies hourly land-surface fields on a 0.1° latitude–longitude grid (Muñoz-Sabater et al., 2021). We use 2-m air temperature, 2-m dew-point temperature and surface pressure for June–August 1991–2025 over 105–125°E and 20–42°N. The rectangle provides a regular support spanning the subtropical-to-temperate north–south gradient of eastern China while excluding most of the Tibetan Plateau west of 105°E. Its northern and southern limits bound that gradient. The analysis panel begins in 1991.

The historical extension covers all 41 June–August seasons from 1950, the first year of the ERA5-Land record, through 1990. It uses the same sites, field construction and five-scale estimator. We report separate summaries for 1950–1978 and 1979–1990 because the earlier segment uses the preliminary ERA5 back extension (ECMWF production documentation). The extension plan and its 99,999-draw product cyclic-shift calculation were recorded before the historical values were retrieved.

Every 16th longitude and 18th latitude cell from a fixed north-west origin gives a 13×13 candidate lattice; removing ocean cells leaves 121 sites (Figure 2). The 35 summers contain 9,350,880 quality-controlled hour–site records and 92 complete UTC fields per summer. Because of NetCDF packing, 2,092 dew-point values exceeded air temperature slightly (0.022% of records); these pairs were clipped at equality before WBT was calculated. The main target is a site average. For area weighting, $a_i = \cos(\text{latitude}_i)$,

$$\bar{y}_t^{(a)} = \frac{\sum_i a_i y_{it}}{\sum_i a_i}, \qquad Q_{h,a}(\mathbf{y}) = \frac{\sum_{i<j} a_i a_j w_{ij,h} (y_i - y_j)^2}{2 \sum_{i<j} a_i a_j w_{ij,h}}.$$

The first area-weighted calculation replaces Q_h by $Q_{h,a}$ using the original labels; the second redefines the labels from $\bar{y}_t^{(a)}$ at the regional peak hours. The two calculations separate spatial weighting from event relabelling.

WBT is calculated from temperature T and dew point T_d in kelvin and surface pressure p in pascals. Vapour pressure is $e(T_d) = 611.2 \exp\{17.67(T_d - 273.15)/(T_d - 29.65)\}$ Pa, and the Bolton mixing-ratio, lifting-condensation-temperature and equivalent-potential- temperature relations are evaluated in these units (Bolton, 1980). Saturated T_w is obtained by solving

$$\theta_e(T_w, T_w, p) = \theta_e(T, T_d, p) \tag{14}$$

at the original pressure with 40 bisection iterations on $[T_d, T]$; even a 100-K initial interval then falls below 10^{-9} K. Values with $T_d > T$ are clipped at equality before evaluation. The Stull approximation provides a comparison (Stull, 2011); the main calculation follows the pseudoadiabatic definition discussed by Davies-Jones (2008).

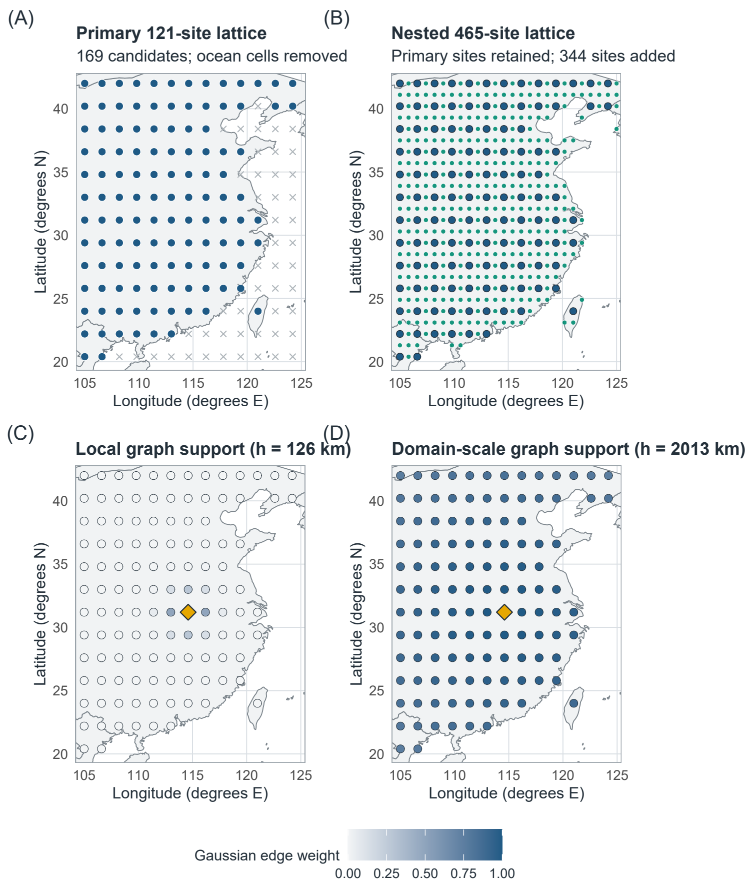

Figure 2 Spatial sampling and graph support. (A) Analysis lattice with 121 land sites from 169 candidates. (B) Nested lattice adding 344 land sites while retaining every analysis site. (C,D) Gaussian edge weights from one reference site under the 126- and 2,013-km graphs, showing how bandwidth changes the geographic support at fixed coordinates. Basemap: Natural Earth 1:50m, accessed through the R `maps` package.

For each UTC day, the 121 simultaneous values at the peak regional mean form the daily field. Within each month-year, the upper regional-mean quartile is high, the middle half is middle and the lower quartile is excluded. Labels depend on regional mean WBT, and the selected hour maximises that mean; graph dispersion enters after selection. UTC+8 days, alternative thresholds, sitewise maxima, all 24 fixed UTC hours and daily-mean fields provide comparisons. At each fixed hour, both the original peak-based labels and labels recomputed from that hour's regional mean are analysed; daily-mean labels are likewise recomputed from daily-mean WBT. A dated event manifest records the type-7 thresholds, membership and absence of ties.

Three checks address within-month seasonal progression: comparison of the day-of-month distributions, removal of a separate linear day-of-month trend at every site within each year–month, and subtraction of each site's calendar-day climatology, estimated from all summers except the one being analysed. The last two calculations use the original event labels. A date-matched calculation compares each high day with all middle days in the same month-year lying within ± 3 or ± 5 calendar days. It first averages the available high-day-specific ratios and then uses the same equal-month, equal-scale and equal-summer hierarchy as the estimator; unmatched-day and empty-record counts are retained.

A continuous diagnostic regresses $\log Q_h$ on centred regional mean WBT within each month-year and bandwidth. Slopes are then averaged equally over months, bandwidths and evaluation summers. This slope supplements the quartile estimand with a graded summary.

Alternative graph kernels are $\exp(-d_{ij}/\rho)$ and $[1-d_{ij}/\rho]_+^2$. For each Gaussian scale, ρ is found by bisection so that the alternative kernel has the same edge-weighted mean pair distance. Across the five scales, their effective edge counts range from 334 to 7,112 and from 382 to 7,113, compared with 366 to 7,109 for the Gaussian graphs; the full matched-distance and effective-edge metadata accompany the analysis.

A nested $0.8^\circ \times 0.9^\circ$ lattice adds 344 land sites while retaining all 121 locations. Numerical convergence is assessed at the five physical bandwidths by comparing 121 sites with a 239-site longitude refinement, a 236-site latitude refinement and all 465 sites under the evaluation events; peak times and labels are also reselected on the full grid. Thin-plate REML surfaces (Wood, 2003) are used only for display in Figures 3 and 4; every estimate uses observed nodes.

We compare ERA5-Land WBT with observations from the National Oceanic and Atmospheric Administration Integrated Surface Database (NOAA ISD) (Smith et al., 2011). The station comparison uses ten summers (1992, 1996, 2000, 2004, 2008, 2012, 2016, 2020, 2023 and 2025) selected before station retrieval. Archive, operating-period and geographic eligibility rules followed by a deterministic maximin design selected 30 dispersed stations. A station-year was retained when it had at least 400 quality-controlled, complete exact-hour observations of temperature, dew point and sea-level pressure; station elevation supplied surface pressure under a standard atmosphere. At two offshore stations, all variables were missing from the nearest ERA5-Land grid cells, leaving 28 usable stations and 175,172 exact-hour WBT pairs.

The station-field calculation uses peak times and high–middle labels from the 121-site ERA5-Land fields. At each retained time, NOAA and ERA5-Land graph contrasts use the identical station subset and require at least 10 stations. All 30 year-month records retained at least one field in each regime. The 503-, 1,006- and 2,013-km graphs were designated as station-supported before evaluation; the two narrower graphs diagnose the limit of the station geometry. The comparison assesses measurement agreement and the direction of spatial contrast on this sparse network.

5.2. Evaluation-period multiscale result

Across the 33 evaluation summers, the five-scale contrast was -7.28%. None of the 99,999 independently sampled product cyclic shifts produced a statistic as negative as the observed value, giving the plus-one one-sided value $p = 0.00001$ under the conditional product-invariance null. This was the minimum attainable value for the Monte Carlo run. The complete five-scale daily profile was shifted jointly within each of the 99 month-year records.

Twenty-eight summer-specific contrasts were negative, and leave-one-summer-out estimates ranged from -7.69% to -6.64%. For process-model comparison, the 95% independent-summer t-reference interval is $[-10.00, -4.57]\%$. The fraction negative is 0.85 (Wilson interval $[0.69, 0.93]$), with a one-sided independent-sign reference value of 3.3×10^{-5}.

The fitted trend was -0.22 percentage points per year (95% descriptive OLS reference interval $[-0.48, 0.05]$, two-sided $p = 0.10$). The 2008–2025 mean was 2.95 points more negative than the 1991–2007 mean, with a descriptive Welch reference interval of $[-8.43, 2.52]$ points. Both intervals include zero, leaving the temporal trend uncertain.

Effects increase in magnitude from -2.86% under the 126-km graph to -13.27% under the 2,013-km graph (Table 3; Figure 3). Negative-summer counts rise from 22 to 30. A 31-point log-spaced bandwidth profile decreases at each successive evaluated bandwidth across the same 126–2,013-km range. Complete-graph variance falls 14.45% and is negative in all 33 summers.

Historical extension, 1950–1990

The 1950–1990 extension gave a five-scale contrast of -11.04% (95% descriptive interval $[-12.84, -9.24]\%$). Forty of 41 summer contrasts were negative, and the leave-one-summer-out range was -11.33% to -10.62%. Its 99,999-draw product-shift calculation produced no statistic as negative as the observed value (plus-one $p = 0.00001$). The scale profile was $-5.96, -7.51, -10.08, -14.50$ and -17.17% from 126 to 2,013 km, with 38, 39, 40, 41 and 41 negative summers. The negative sign and strengthening contraction with bandwidth also appear in the 41 earlier summers.

The production-history split gave -10.50% in 1950–1978 (28 of 29 summers negative) and -12.34% in 1979–1990 (12 of 12 negative). The segments are reported separately because the pre-1979 record uses different forcing. The historical climatology–anomaly decomposition re-estimates the site-specific monthly climatological field from the 41

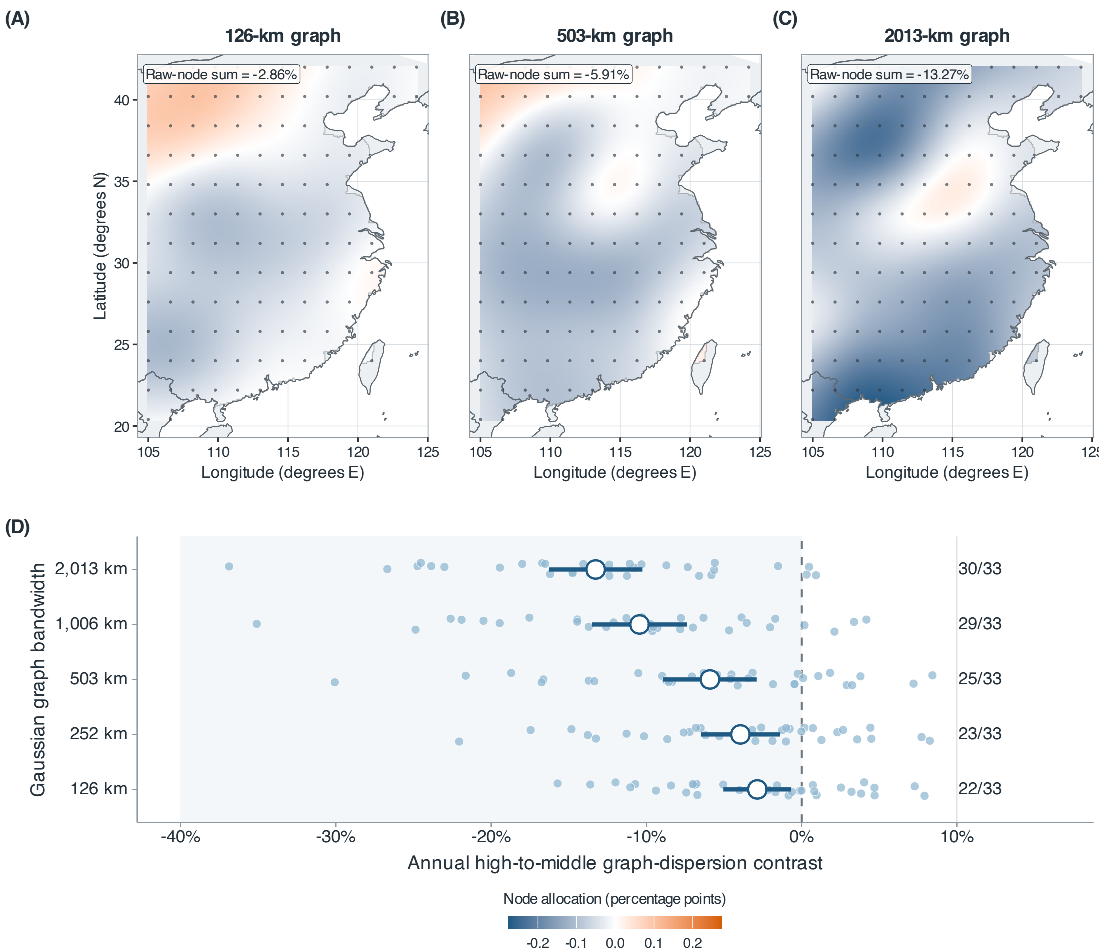

Figure 3 Scale-specific node allocations and summer contrasts for the 33 evaluation summers. (A–C) Allocations for the 126-, 503- and 2,013-km graphs share a symmetric colour scale; their observed-node sums are -2.86%, -5.91% and -13.27%. The interpolated surfaces visualise the 121 observed-node values; estimation and inference use the observed nodes. (D) Small circles show summer-specific contrasts at all five bandwidths; large circles and bars show means and 95% descriptive intervals. Labels give the numbers of negative summers.

summers in 1950–1990. At 2,013 km, the historical anomaly-energy component was -0.28 points (95% descriptive interval $[-1.01, 0.45]$), whereas the climatology–anomaly cross component was -16.89 points ($[-19.31, -14.47]$) and negative in all 41 summers. Both the latitude and latitude–longitude structured components were negative in every summer at every bandwidth. In the earlier record, most broad-scale contraction was likewise in the alignment term. The 1950–1978 estimates inherit the uncertainty of the preliminary forcing data.

5.3. Spatial structure and energy decomposition

The decomposition in Equation (10) attributes most of the contrast to the north–south gradient. At 126 km it accounts for -2.58 points (95% independent-summer reference interval $[-2.92, -2.24]$) of the -2.86-point contrast. From 252 to 2,013 km its mean contribution is more negative than the total ($-4.60, -8.38, -12.73, -14.96$ points), while residual contrasts contribute $+0.66, +2.48, +2.29, +1.69$ points. The gradient component is negative in all 33 evaluation summers at every bandwidth. Adding longitude to the basis makes the 2,013-km structured component -16.23 points, compared with -14.96 points for latitude alone; the corresponding residuals are positive. The

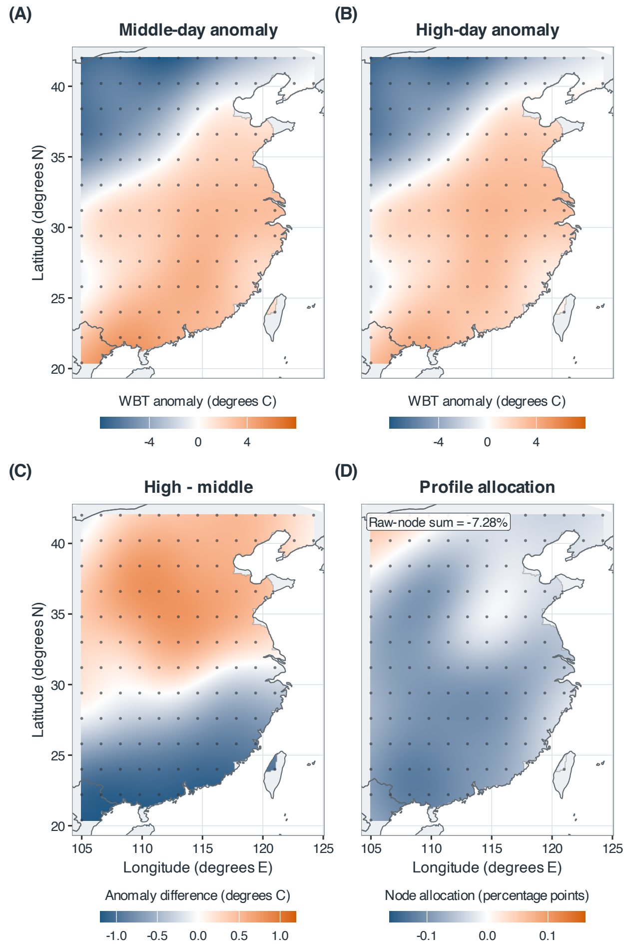

Figure 4 Spatial structure of the 121-site evaluation contrast. (A,B) Middle- and high-day spatially centred fields; (C) their difference; (D) exact observed-node allocations averaged over the five graph scales. The 121 allocations sum to -7.28%, identical to the five-scale estimand. The interpolated surfaces visualise the 121 observed-node values; estimation and inference use the observed nodes.

latitude–longitude–elevation basis gives a 2,013-km structured component of -14.34 points (95% descriptive interval $[-16.75, -11.92]$), negative in all 33 summers, and a residual of $+1.07$ points. At 126 km, however, its structured component is -1.16 points ($[-2.66, 0.33]$) and is negative in only 18 summers. At broad scales, the nested bases show that latitude accounts for most of the low-dimensional geographic–topographic structure; structured attribution is weaker at 126 km. Node allocations are negative at 106 sites and sum exactly to -7.28%. Each is a symmetric incident-edge allocation.

The climatology–anomaly decomposition in Equation (11) locates the broad-scale change. At 126 km, anomaly-energy change contributes -2.63 points and the climatology–anomaly cross term contributes -0.22 points. At 2,013 km, the anomaly-energy component is $+0.41$ points (reference interval $[-0.50, 1.31]$), whereas the cross term is -13.68 points ($[-16.68, -10.67]$) and is negative in all 33 summers. High-day anomalies therefore tend to oppose the persistent broad geographic contrast, while their own broad-scale energy changes little.

The convergence analysis compares each grid with the 465-site result using the evaluation events. The 121-site scale profile differs by at most 0.99 percentage points, with a five-scale root mean squared difference of 0.76 points. Latitude refinement reduces these values to 0.50 and 0.38 points; longitude refinement gives 0.93 and 0.73 points. Adding all sites with the evaluation events yields -8.03%; reselecting events yields -7.25%. Contraction strengthens

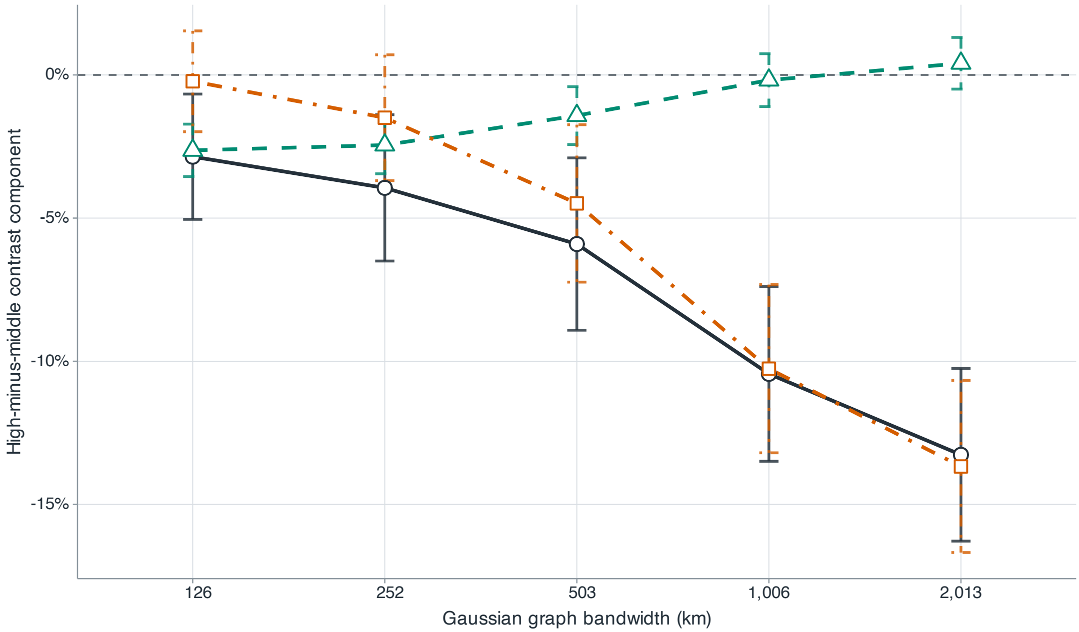

Figure 5 Exact decomposition of the raw-field graph contrast. Dark circles, teal triangles and orange squares denote total change, anomaly-energy change and the climatology–anomaly cross component; bars are 95% descriptive intervals. Monthly-climatology energy cancels within each record.

with bandwidth at every resolution. Agreement between spatial supports is weakest at 126 km, where the 121-site spacing and station support are coarsest.

The graph effects are relative changes in weighted squared differences. The five-scale mean corresponds to a -3.71% RMS change when transformed as $\sqrt{1-0.0728}-1$; averaging record-specific RMS ratios gives -3.97%.

Threshold, Stull and sitewise-maximum sensitivities range from -6.90% to -8.27%. Early- and late-period summaries have overlapping descriptive intervals. These checks assess the stability of the spatial contrast across analysis choices. For comparison, the same equal-month, equal-scale and equal-year formula gives -15.14% for the 2015 and 2022 design summers.

5.4. Robustness and external measurement agreement

The 495 middle-day denominators range from 2.25 to $25.99\ ^\circ\mathrm{C}^2$ across records and scales; within-scale coefficients of variation are 0.076–0.177, with a minimum of $2.25\ ^\circ\mathrm{C}^2$. The raw ratio is numerically stable over these data; the heavy-tail simulation examines its behaviour with small denominators. The back-transformed log-ratio is -8.30%; the bounded symmetric contrast $2(Q^H-Q^M)/(Q^H+Q^M)$ is -8.59%. Cosine-latitude area weighting gives -8.11% using the original labels, and -7.67% when labels are recomputed from the area-weighted regional mean. Kernels matched to the Gaussian weighted mean distance give -7.71% (exponential) and -7.33% (compact quadratic). All preserve the negative direction and strengthening contraction with bandwidth.

The continuous-intensity diagnostic gives a five-scale $\log Q_h$ slope of $-0.065\ ^\circ\mathrm{C}^{-1}$ (95% descriptive interval $[-0.082,-0.048]$), with negative summer slopes in 31 of 33 years. Scale-specific slopes become more negative from $-0.032\ ^\circ\mathrm{C}^{-1}$ at 126 km to $-0.110\ ^\circ\mathrm{C}^{-1}$ at 2,013 km; the corresponding negative-year counts rise from 29 to 32. The negative association extends across the within-month WBT distribution.

The estimated contrast depends on the definition of the spatial field. Subtracting each site's 35-year monthly climatology reduces the estimate to -3.92% ($[-8.67, 0.83]\%$); subtracting the site-year-month mean gives -1.32% ($[-5.63, 2.99]\%$). Site-month standardisation yields -9.87%, but targets a different dimensionless field. Together with the gradient decomposition, these results show that high-day anomalies offset the cross-site climatological gradient, while their own broad-scale energy changes little.

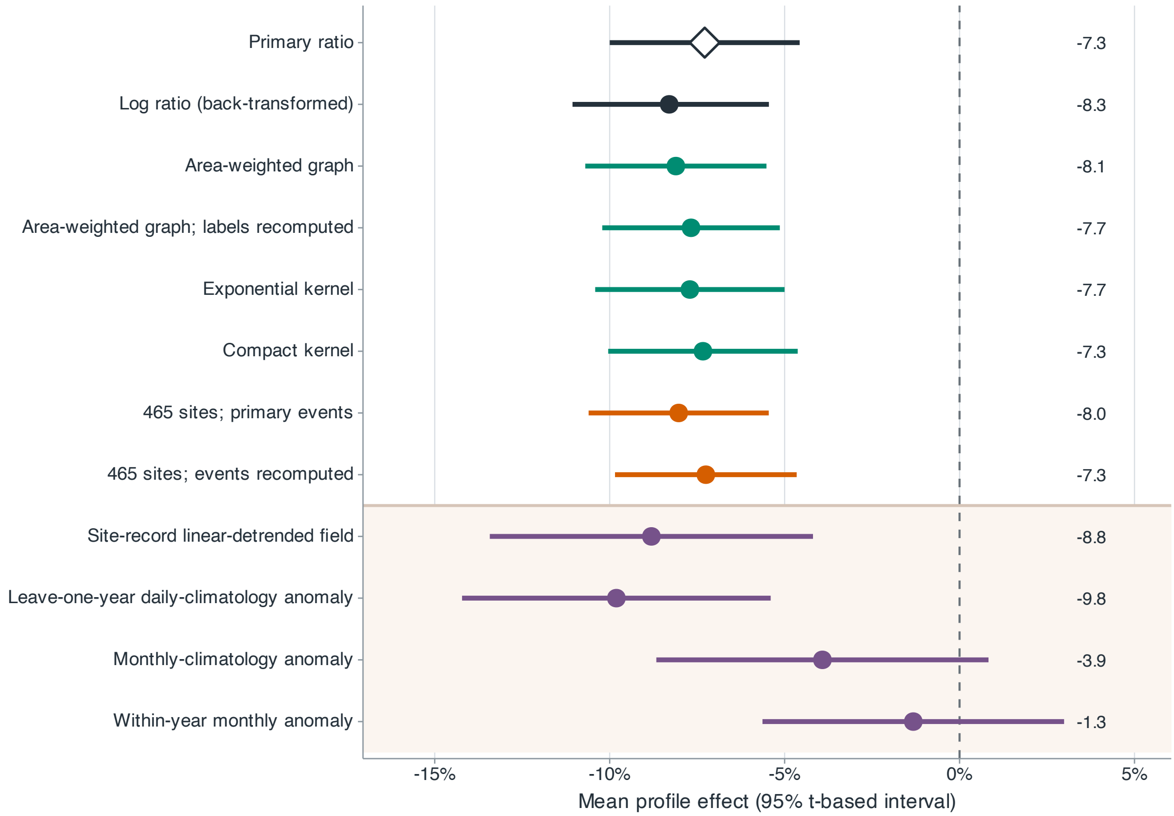

Figure 6 Robustness of the mean profile effect. Dark rows denote the main and transformation analyses; green, orange and purple rows denote graph construction, spatial resolution and field definition. Points and bars are estimates and 95% descriptive intervals.

The regime dates have a systematic but nonmonotone seasonal pattern. Averaged over June–August, high days occur 1.96 days later than middle days (95% t-reference interval $[1.10, 2.83]$ days): they occur 7.82 and 6.34 days later in June and July, but 8.29 days earlier in August. Removing a separate linear day-of-month trend within each site–year–month gives -8.81% ($[-13.43, -4.19]\%$); subtracting the leave-one-year calendar-day climatology gives -9.81% ($[-14.22, -5.40]\%$). The negative contrast persists after both seasonal adjustments, although the weaker site-month and site–year–month anomaly results show that different demeaning operations target different spatial fields.

Calendar matching attenuates the effect but preserves its direction. With 95% descriptive intervals in parentheses, the ± 3-day estimator is -5.21% ($[-8.01, -2.41]\%$), with 24 of 33 summer effects negative and 670 of 792 high days matched; the ± 5-day estimator is -5.72% ($[-8.43, -3.02]\%$), with 25 negative summers and 750 matched high days. Every one of the 99 month-year records has at least one eligible match under both windows. The profiles run from -2.32% to -8.87% for ± 3 days and from -2.54% to -9.89% for ± 5 days, retaining the strengthening contraction with bandwidth.

All fixed-hour and daily-mean analyses yielded negative estimates. Using the original event days, all 24 fixed-hour estimates are negative, from -10.07% at 00 UTC to -6.03% at 11 UTC, and daily-mean fields give -7.58%. When labels are recomputed from each fixed hour, all estimates remain negative, from -14.65% at 22 UTC to -5.42% at 11 UTC; labels recomputed from daily-mean WBT give -9.14%. Peak hours have nearly identical distributions in the original high and middle groups (median 06 UTC). Figure 6 compares the main estimate with transformation, graph-construction, spatial-resolution and field-definition alternatives.

The station comparison contains 175,172 exact-hour matches at 28 usable stations. ERA5-Land minus station WBT has a bias of -0.263 °C, a mean absolute error of 0.938 °C and a root mean squared error of 1.270 °C. The

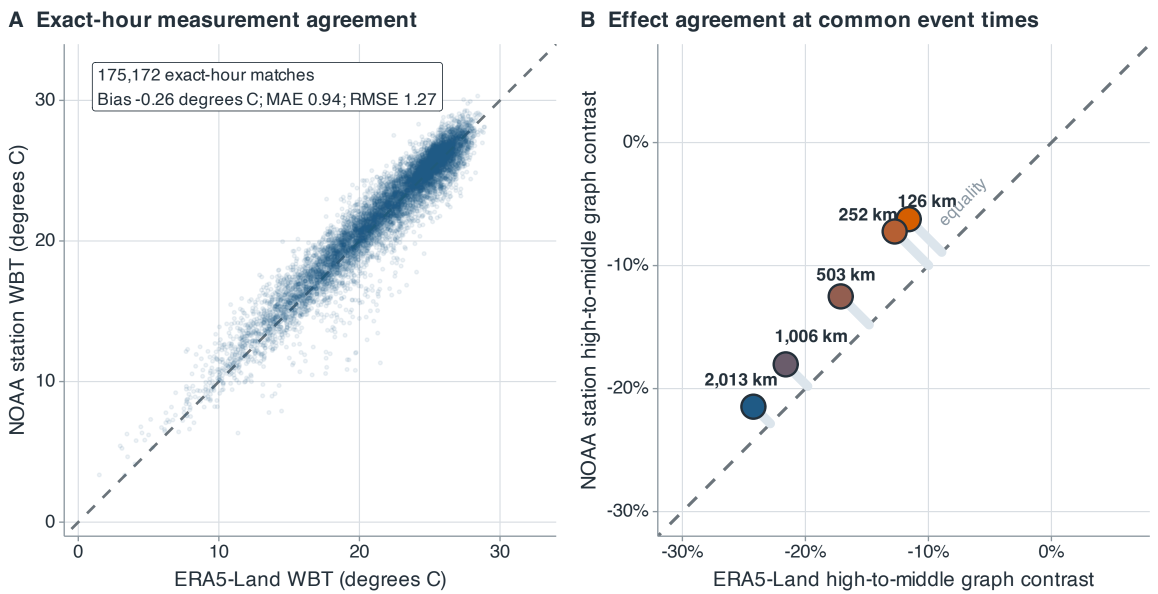

Figure 7 NOAA station and ERA5-Land agreement in ten station-comparison summers. (A) Exact-hour WBT at 28 stations, with summaries from all 175,172 matches. (B) High–middle graph contrasts at the five bandwidths, using ERA5-Land peak times and labels and identical station subsets. Dashed diagonals denote equality; pale segments in panel B connect station estimates to equality.

within-station centred correlation is 0.916. The two offshore stations excluded from these summaries had no finite values at their nearest ERA5-Land grid cells.

At 126, 252, 503, 1,006 and 2,013 km, the station high–middle contrasts are $-6.23, -7.23, -12.50, -18.03$ and -21.47%; the corresponding ERA5-Land values on identical station subsets are $-11.60, -12.74, -17.13, -21.60$ and -24.23%. The numbers of negative station summers across the five scales are 6, 7, 8, 9 and 9. Averaged over the three predesignated station-supported scales, the station effect is -17.34% ($[-28.49, -6.18]\%$), with 9 of 10 summer contrasts negative; the paired ERA5-Land effect is -20.99% ($[-28.57, -13.40]\%$), with all 10 negative. The station and matched reanalysis estimates agree in direction at the three designated broad scales. At 126 km, both intervals include zero and provide no clear evidence of a local-scale contrast (Figure 7).

6. Discussion

Across the 33 evaluation summers, spatial WBT contrast was lower when the regional mean was high. The five-scale mean was -7.28%, 28 summer effects were negative, and the product cyclic-shift assessment gave $p = 0.00001$ under the conditional product-invariance null. Contraction strengthened from 126 to 2,013 km, with the largest estimated contraction at broad geographic scales. The 1950–1990 extension showed the same scale pattern, with a mean of -11.04% and 40 of 41 summer effects below zero. Station data also agreed in direction at 503 km and broader: the three-scale mean was -17.34%, with negative contrasts in 9 of 10 summers. At 126 km, the station and matched reanalysis intervals both included zero. These contrasts describe spatial arrangement, a feature distinct from the area exceeding a heat threshold.

Our analysis integrates scale-indexed graph dispersion, finite-record randomisation and exact spatial accounting for an observational field. The five graphs define a single estimand while retaining its scale profile, product shifts preserve within-record dependence among scales, and node and energy decompositions recover the regional contrast exactly. Together, these elements show how the change is distributed over nodes and scales. The finite-record calculation and stochastic process model answer different inferential questions. Product-shift rejection remained near nominal in the simulated null settings. Separate simulations show why Student and HAC intervals can be unreliable with only 33 dependent summers.

The decompositions locate the contrast in the spatial field. After removing each day's regional mean, high-day fields are relatively warmer in the north and cooler in the south than middle-day fields, and the north–south component accounts for most of the contrast. At broad scales, the climatology–anomaly cross component accounts for nearly all of the contraction, whereas the anomaly field's own energy changes little. High regional mean WBT is therefore associated with a flatter relative geographic pattern. The negative raw-field contrast persists under area weighting, alternative kernels, fixed-hour analyses and calendar matching. Its attenuation after site-month and site–year–month demeaning further locates the result in the cross-site climatic gradient.

Several limitations remain. The 126-km graph has the coarsest effective spatial support and the weakest station evidence. The rectangular domain defines the sampling support, and other boundaries may produce different allocations. The historical extension uses the same reanalysis product, and the 1950–1978 segment carries the documented preliminary-forcing caveat. The station comparison covers 28 locations in ten summers and uses event times and labels from ERA5-Land. It measures agreement for those events on the sampled station network. Pressure-level circulation, population-weighted fields and health outcomes are needed to examine mechanism and exposure. Across the analysed finite record, high-regional-mean days had weaker broad-scale WBT contrast while transient anomaly energy changed little. Generalisation to a long-run climate process requires longer or independent records.

Data availability

ERA5-Land is available from the Copernicus Climate Data Store (`https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land`). The NOAA ISD station archive (Smith et al., 2011) is available at `https://www.ncei.noaa.gov/products/land-based-station/integrated-surface-database`. The processed data, site and station manifests, quality-control records, derived results and complete analysis code are available in the public GitHub repository `https://github.com/Stork343/eastern-china-wet-bulb-contraction`. On acceptance, a versioned release will be archived with a persistent identifier. The raw reanalysis and station archives remain available through their source providers.

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Conflict of interest

The authors declare no conflict of interest.

References

Andrews, D.W.K. (1991). Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59, 817–858.

Bolton, D. (1980). The computation of equivalent potential temperature. *Monthly Weather Review*, 108, 1046–1053.

Chagnaud, G., Taylor, C.M., Jackson, L.S., Birch, C.E., Marsham, J.H. and Klein, C. (2025). Wet-bulb temperature extremes locally amplified by wet soils. *Geophysical Research Letters*, 52, e2024GL112467.

Cressie, N.A.C. (1993). *Statistics for Spatial Data*, revised edn. New York: Wiley.

Davies-Jones, R. (2008). An efficient and accurate method for computing the wet-bulb temperature along pseudoadiabats. *Monthly Weather Review*, 136, 2764–2785.

Dong, X., Thanou, D., Frossard, P. and Vandergheynst, P. (2016). Learning Laplacian matrix in smooth graph signal representations. *IEEE Transactions on Signal Processing*, 64, 6160–6173.

Geary, R.C. (1954). The contiguity ratio and statistical mapping. *The Incorporated Statistician*, 5, 115–145.

Healy, D., Tawn, J., Thorne, P. and Parnell, A. (2025). Inference for extreme spatial temperature events in a changing climate with application to Ireland. *Journal of the Royal Statistical Society: Series C*, 74, 275–299. doi:10.1093/jrsssc/qlae047.

Harris, K.D. (2020). A shift test for independence in generic time series. *arXiv:2012.06862*.

Horton, D.E., Johnson, N.C., Singh, D., Swain, D.L., Rajaratnam, B. and Diffenbaugh, N.S. (2015). Contribution of changes in atmospheric circulation patterns to extreme temperature trends. *Nature*, 522, 465–469.

Ibragimov, I.A. and Linnik, Y.V. (1971). *Independent and Stationary Sequences of Random Variables*. Groningen: Wolters-Noordhoff.

Moran, P.A.P. (1950). Notes on continuous stochastic phenomena. *Biometrika*, 37, 17–23.

Muñoz-Sabater, J. et al. (2021). ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data*, 13, 4349–4383.

Newey, W.K. and West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55, 703–708.

Pepin, N.C. et al. (2015). Elevation-dependent warming in mountain regions of the world. *Nature Climate Change*, 5, 424–430.

Phipson, B. and Smyth, G.K. (2010). Permutation p-values should never be zero: calculating exact p-values when permutations are randomly drawn. *Statistical Applications in Genetics and Molecular Biology*, 9, Article 39.

Raymond, C., Matthews, T. and Horton, R.M. (2020). The emergence of heat and humidity too severe for human tolerance. *Science Advances*, 6, eaaw1838.

Seneviratne, S.I., Corti, T., Davin, E.L., Hirschi, M., Jaeger, E.B., Lehner, I., Orlowsky, B. and Teuling, A.J. (2010). Investigating soil moisture–climate interactions in a changing climate: a review. *Earth-Science Reviews*, 99, 125–161.

Sherwood, S.C. and Huber, M. (2010). An adaptability limit to climate change due to heat stress. *Proceedings of the National Academy of Sciences*, 107, 9552–9555.

Shreevastava, A., Raymond, C. and Hulley, G. (2023). Contrasting intraurban signatures of humid and dry heatwaves over Southern California. *Journal of Applied Meteorology and Climatology*, 62, 709–720.

Shuman, D.I., Narang, S.K., Frossard, P., Ortega, A. and Vandergheynst, P. (2013). The emerging field of signal processing on graphs: extending high-dimensional data analysis to networks and other irregular domains. *IEEE Signal Processing Magazine*, 30, 83–98.

Simmons, J.P., Nelson, L.D. and Simonsohn, U. (2011). False-positive psychology: undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, 22, 1359–1366.

Smith, A., Lott, N. and Vose, R. (2011). The integrated surface database: recent developments and partnerships. *Bulletin of the American Meteorological Society*, 92, 704–708. doi:10.1175/2011BAMS3015.1.

Speizer, S., Raymond, C., Ivanovich, C. and Horton, R.M. (2022). Concentrated and intensifying humid heat extremes in the IPCC AR6 regions. *Geophysical Research Letters*, 49, e2021GL097261.

Stull, R. (2011). Wet-bulb temperature from relative humidity and air temperature. *Journal of Applied Meteorology and Climatology*, 50, 2267–2269.

White, H. (2000). A reality check for data snooping. *Econometrica*, 68, 1097–1126.

Wood, S.N. (2003). Thin plate regression splines. *Journal of the Royal Statistical Society: Series B*, 65, 95–114.

Yuan, A.E. and Shou, W. (2024). A rigorous and versatile statistical test for correlations between stationary time series. *PLOS Biology*, 22, e3002758.

Table 1. Data-generating processes in the simulation study. Day-level fields use the observed 121-site coordinates, and all metrics are computed from identical simulated fields and joint shifts.

Experiment	Component	Specification
Day-level	Baseline field	Gaussian anomaly field projected to have exactly zero spatial sample mean; exponential spatial covariance with 450-km range and 5% nugget; temporal coefficient 0.45; six records of 30–31 days.
	Event labels	Regional-mean series, autoregressive with coefficient 0.65 and equal to the simulated field mean; centred anomalies are independent of this series; high and middle labels use within-record quartiles.
	Null variants	One aspect varied at a time: three or 12 records; 20 or 60 days per record; temporal coefficients 0 or 0.75; independent t_3 radial scaling; two non-circular finite autoregressions with coefficients 0.45 and 0.75.
	Alternatives	Anomaly standard deviation $\times 0.97, 0.955, 0.94, 0.92, 0.90$; correlation range 525, 587, 650, 725, 800 km; gradient amplitude 1.6, 1.5, 1.4, 1.3, 1.2 standardised units (baseline 1.8).
	Replication	25 design cells $\times$ 1,000 data sets; 499 joint product shifts per test.
Summer-level	Yearly effects	Autoregressive coefficients 0, 0.3, 0.6; marginal standard deviation 0.12; innovations Gaussian, centred log-normal or standardised t_3; 200-year burn-in.
	Sample	$R = 20, 33, 60, 120$ summers; true mean effects $0, -3.5\%, -7\%$; 108 cells $\times$ 10,000 replications.

Table 2. Power comparison over all nine spatial alternatives. Entries after the true graph-profile effect are rejection rates (%) from 1,000 simulated data sets. Moran's I and Geary's C use two-sided tests; the five dispersion procedures use lower-tail tests.

Mechanism	Strength	True effect (%)	Graph profile	Five-bin variogram	Variance	Nearest neighbour	Single-bin semivariance	Moran I	Geary C
Amplitude	Weak	-5.91	50.0	33.6	31.1	76.4	45.0	5.2	5.9
	Medium	-11.64	95.3	79.8	77.1	99.8	91.9	4.6	4.5
	Strong	-19.00	99.9	99.3	99.1	100.0	99.9	5.4	4.7
Range	Weak	-6.07	48.9	19.0	16.4	95.4	51.5	15.5	15.9
	Medium	-14.38	98.6	62.4	53.3	100.0	99.1	45.3	53.7
	Strong	-22.20	100.0	91.4	85.6	100.0	100.0	74.0	85.0
Gradient	Weak	-10.34	90.9	93.9	96.5	40.8	78.9	32.4	81.6
	Medium	-19.47	100.0	100.0	100.0	82.5	99.8	87.6	100.0
	Strong	-27.37	100.0	100.0	100.0	97.6	100.0	99.8	100.0

Table 3. Physical scale of the evaluation-period graph contrasts. Q^M and Q^H are equal-record descriptive means in $^\circ\mathrm{C}^2$; RMS columns report $(2Q)^{1/2}$ in $^\circ\mathrm{C}$. The graph effect is the mean of record-specific squared-difference ratios, so it need not equal the ratio of the two descriptive Q columns. Negative gives the number of summers with a negative graph effect.

Bandwidth (km)	Q^M ($^\circ$C^2)	Q^H ($^\circ$C^2)	RMS^M ($^\circ$C)	RMS^H ($^\circ$C)	Graph effect (%)	Negative (/33)
126	2.66	2.58	2.31	2.27	-2.86	22
252	4.36	4.17	2.95	2.89	-3.94	23
503	7.94	7.43	3.99	3.86	-5.91	25
1,006	12.89	11.49	5.08	4.79	-10.44	29
2,013	15.90	13.74	5.64	5.24	-13.27	30